

Confusion Matrix

		Predicted Class	
		Predicted Positive	Predicted Negative
Actual Class	Actual Positive	True Positive (TP) <i>Correct Prediction</i>	False Positive (FP) <i>Type I Error (False Alarm)</i>
	Actual Negative	False Negative (FN) <i>Type II Error (Missed)</i>	True Negative (TN) <i>Correct Rejection</i>

 = Correct  = Incorrect

Inquiry into Model Performance from a Non-Technical Perspective:

1. Assessment of overall model quality?
Accuracy
2. Did the model correctly identify actual buyers?
Recall (Purchased)
3. Were non-buyers wrongly targeted?
Precision-related metric
4. Is there a high degree of confidence in positive predictions?
Precision
5. Is the model suitable for deployment in a production environment?
F1 Score

Inquiry into Model Performance from a Technical Perspective:

1. What is the True Positive Rate?
Recall / Sensitivity
2. How many False Positives exist?
From Confusion Matrix FP value
3. What is the Precision for positive class?
Precision = $TP / (TP + FP)$
4. Is the dataset imbalanced? Which metric is reliable?
Use F1 Score / Recall instead of Accuracy
5. What is the Type-II error rate?
False Negative Rate